

UCS503 - Software Engineering

CLOTH TRACER

A Smart Laundry Management System with AI-Assisted Cloth Matching

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Introduction

1.1 Background and Context

Laundry services are commonly used by students living in hostels and campuses. However, traditional laundry management often involves manual order placement, physical receipts, and limited communication about the status of clothes. Students may have difficulty knowing whether their laundry has been received, is being processed, or is ready for collection.

Cloth Tracer is a smart laundry management system designed to digitize this process. It provides a centralized platform where students can place laundry orders, view order status, receive notifications, and submit complaints. The system also provides laundry staff with tools to manage orders and respond to complaints, making the overall process more organized and transparent.

1.1.1 Problem Statement

Traditional laundry management systems often rely on manual record-keeping and physical communication between students and laundry staff. This can lead to difficulties in tracking order status, maintaining accurate records, and communicating updates. Students may also face challenges when reporting complaints or following up on unresolved issues. Misplacement of clothes during the laundry process is a significant challenge, particularly when multiple students' clothes are being processed simultaneously. In the absence of proper cloth records, identifying and locating a misplaced item can be difficult. Cloth Tracer addresses this issue by capturing photographs of clothes at the time of laundry submission and associating them with the corresponding laundry order. These photographs serve as a visual reference for identifying missing clothes and support the complaint-handling process. Students can report misplaced clothes through the complaint system, while laundry staff can review the associated clothing information to assist in identification and resolution.

1.2 Project Objectives

1. **Objective 1:** To develop a centralized web-based laundry management system that enables students to submit laundry requests and access their digital laundry records and receipts through a single platform.
2. **Objective 2:** To maintain photographic records of submitted clothes and as-

sociate them with individual laundry items, enabling visual reference during missing or damaged clothing complaints and supporting AI-assisted matching.

3. **Objective 3:** To provide an organized complaint management and notification system that enables students to report laundry-related issues, track complaint status, and receive updates from laundry staff.

Methodology and Design

2.1 System Architecture

Cloth Tracer is designed as a centralized web-based laundry management system that supports two primary roles: Student/User and Admin. The system follows a client-server architecture in which the frontend provides the user interface, the backend handles application logic and communication, and the database stores structured application data.

Students interact with the system through the frontend to register, submit laundry orders, upload garment photographs, view digital receipts, access laundry history, submit complaints, and receive notifications. Administrators use the same platform to manage student records, review laundry submissions, handle complaints, and communicate important updates.

The system uses Firebase Authentication for user identity management, MySQL for structured data storage, and Cloudinary for storing clothes images and complaint-related images. These components work together to provide a centralized and organized laundry management solution.

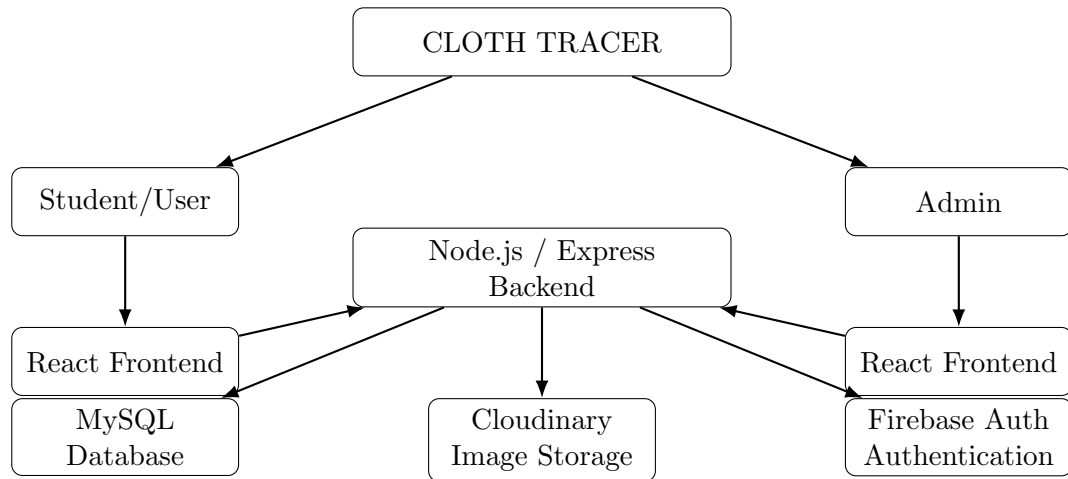


Figure 2.1: High-Level System Architecture of Cloth Tracer

2.1.1 High-Level Design

The high-level design of Cloth Tracer is based on two interconnected user roles: Student/User and Admin. Each role has a separate interface and set of functionalities

according to its responsibilities.

The Student/User module is designed to simplify the laundry submission process. After authentication, students can access their dashboard, submit laundry containing up to ten clothes, upload photographs, select clothes types, and receive a digital receipt. The dashboard also provides access to laundry history, complaints, and notifications.

The Admin module is designed to support centralized laundry management. Administrators can view student records, access laundry submissions, review complaints, and send notifications. The system therefore provides a common platform for both students and administrators while maintaining role-specific access to functionalities.

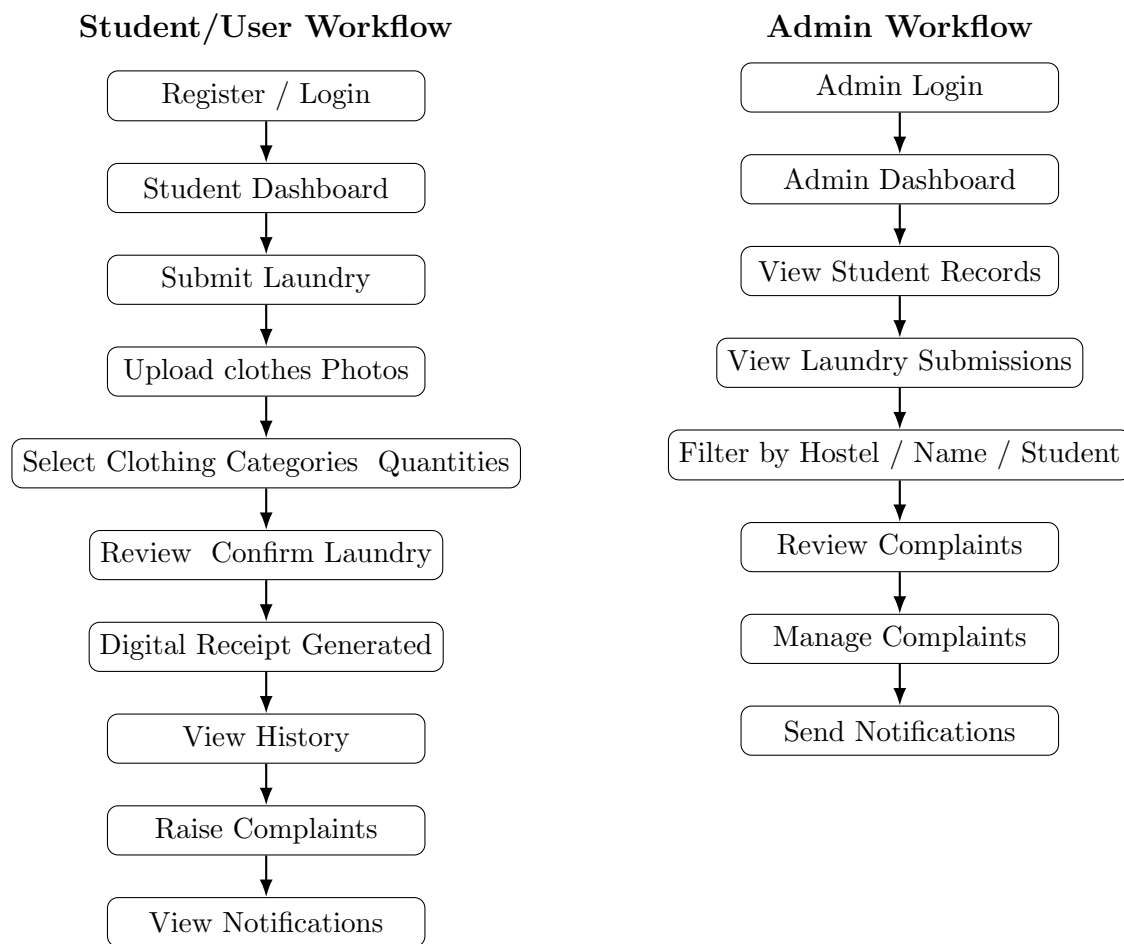


Figure 2.2: Student and Admin Workflows of Cloth Tracer

2.1.2 Technical Stack

- **Framework/Language:** React.js is used for developing the frontend and creating the user interface. Node.js is used for backend development.
- **Database:** MySQL is used to store student details, laundry submissions, cloth information, complaints, and notifications.
- **Tools:** Visual Studio Code is used as the primary development environment for coding, testing, and debugging.
- **Libraries/Services:** Firebase Authentication is used for user authentication and Cloudinary is used for storing cloth and complaint-related images.

2.1.3 Implementation Details

Module 1: Core Functionality

The core functionality of Cloth Tracer is implemented through the Student/User module. Students can register and log in using their personal and academic details, including roll number, department/program, hostel, and personal laundry number.

After logging in, students can access the dashboard and submit a laundry order containing up to ten clothes. For each cloth, the student uploads one photograph and selects the appropriate cloth type, such as T-shirt, towel or other supported clothing categories. The submitted details are stored along with the student's laundry order.

After successful submission, the system generates a digital receipt containing the student's details and the information of the submitted clothes. Students can also view their previous laundry submissions and receipts through the History section.

The stored cloth photographs provide a visual reference for identifying a misplaced cloth. Students can use the available submission details while reporting a laundry-related issue through the complaint management system.

Module 2: Secondary Features

The secondary features of Cloth Tracer include complaint management, notifications, and administrative operations.

The complaint management module allows students to report issues such as misplaced or damaged clothes. When a complaint concerns a specific clothing item, the student can select the relevant laundry item and reuse its original submitted photograph as supporting evidence.

The notification module enables administrators to send important updates to students, such as changes in laundry submission or collection schedules and other service-related announcements.

The Admin module allows administrators to view and filter student and laundry records by hostel, student name , rollno or laundry slip number. Administrators can inspect laundry submissions, review complaints, update complaint status, and manage overall laundry-related records. These features help maintain organized records and improve communication between students and laundry staff.

Results and Evaluation

3.0.1 Testing Methodology

Cloth Tracer was tested to verify the correctness, functionality, and reliability of its major modules. Testing was performed on the Student/User and Admin modules to ensure that the system works as expected.

- **Unit Testing:** Individual features such as registration, login, laundry submission, cloth photo upload, receipt generation, complaint submission, and notification display were tested separately.
- **Integration Testing:** The interaction between the frontend, backend, authentication service, database, and image storage service was tested to ensure that data was transferred and stored correctly.
- **Performance Testing:** The system was checked for response time, page loading, form submission, and image upload performance under normal usage conditions.
- **User Testing:** The system was evaluated by using the main workflows of students and administrators. Feedback was considered to improve the usability and clarity of the application.

3.0.2 Performance Metrics

The performance of Cloth Tracer was evaluated by testing its main functionalities. The evaluation focused on the successful working of student and admin features, proper storage of submitted information, and correct display of records.

The evaluation results show that the major functionalities of Cloth Tracer worked successfully. Students were able to submit laundry orders, upload cloth photographs, view digital receipts, check their laundry history, and raise complaints. Administrators were able to view and manage student records, review complaints, and send notifications.

3.0.3 Analysis

Meeting the objectives. The primary objectives of Cloth Tracer were achieved. The system provides a centralized platform for students to register, submit laundry orders, upload cloth photographs, generate digital receipts, view laundry history, and

Table 3.1: Performance Results Summary

Metric	Achieved Result
Registration and Login	Successful
Laundry Submission	Successful
Cloth Photo Upload	Successful
Digital Receipt Generation	Successful
Laundry History	Successful
Complaint Submission	Successful
Admin Record Management	Successful
Notification Display	Successful
Admin Dashboard —	Successful
Laundry Schedule Management	Successful
Student	Hostel Management
Successful	
Complaint Resolution	Successful
Reports	Analytics
Successful	
Admin Laundry History	Successful

report laundry-related complaints. The Admin module also supports the management of student records, laundry submissions, complaints, and notifications.

Exceeding expectations. The system provides additional organization and convenience by maintaining cloth photographs along with laundry orders. These photographs act as a visual reference when a cloth is misplaced and support the complaint resolution process. The digital receipt and history features also allow students to maintain and access their previous laundry records easily.

Challenges encountered. During development, challenges were encountered in connecting the frontend with the backend, managing user authentication, storing cloth photographs, and maintaining accurate laundry records. These challenges required repeated testing and debugging of the system's different modules.

Resolution of challenges. The development challenges were resolved through continuous testing and debugging of each module. Frontend and backend communication issues were solved by checking API requests and responses. Authentication-related problems were resolved by validating user credentials and testing the login and registration flow. Cloth photo upload issues were addressed by checking file validation and image storage integration. Database-related errors were resolved by validating the submitted data and maintaining proper links between students, laundry orders, clothes, and complaints. These steps helped ensure that the major features of Cloth Tracer worked correctly.

Overall, the evaluation indicates that Cloth Tracer improves laundry record man-

agement, simplifies the complaint process, and supports better communication between students and laundry administrators.

Conclusion

4.1 Summary of Work

Cloth Tracer was developed as a centralized web-based laundry management system for students and laundry administrators. The system allows students to register and log in, submit laundry orders containing up to ten clothes, upload cloth photographs, generate digital receipts, view laundry history, raise complaints, and receive notifications. The Admin module allows administrators to view student and laundry records, filter submissions, manage complaints, and send important updates. The project achieved its main objectives by improving laundry record management, maintaining cloth-related information, and simplifying the complaint-handling process.

4.2 Key Findings

- **Finding 1:** Cloth Tracer successfully centralizes student details, laundry submissions, cloth photographs, receipts, and complaint records in one platform.
- **Finding 2:** Storing cloth photographs with laundry orders provides a useful visual reference for identifying misplaced clothes and supporting complaint resolution.
- **Finding 3:** The digital receipt, laundry history, admin record management, and notification features improve the organization of laundry activities and communication between students and laundry administrators.

4.2.1 Future Work and Recommendations

Potential Enhancements. The system incorporates an AI-assisted visual matching module to help identify potentially misplaced clothing items. An unidentified clothing photograph is compared with previously submitted laundry photographs using visual feature representations and image similarity. The system returns the most probable matching items along with similarity scores. The administrator reviews and verifies the suggested match before confirming the associated student and laundry record.

Alternative Approaches Worth Exploring. Explore advanced Computer Vision / Deep Learning models for clothing recognition. Use image embeddings and similarity models instead of traditional image classification. Integration of QR/ RFID-based clothing identification for large-scale laundry operations. **Scalability Improvements.** The system can be optimized to support thousands of students and a

large number of cloth photographs. MySQL queries and Cloudinary image management can be improved for faster data access and efficient storage. The system can also be extended for deployment across multiple hostels and campuses.

Integration with Other Systems. Cloth Tracer can be integrated with the college ERP or Student Information System to reduce repeated data entry and maintain consistent student records. Integration with the hostel management system can improve coordination between students, hostel authorities, and laundry administrators. Email, SMS, or push notification services can also be added for timely updates. Online payment systems may be integrated in the future if additional paid laundry services are introduced.

Real-World Deployment Considerations. For real-world deployment, the system should include proper data security, user access control, regular database backups, and reliable image storage. The application should also be tested under a large number of users to ensure stability, scalability, and consistent performance.

4.2.2 Final Remarks

The development of Cloth Tracer provided practical experience in frontend development, backend integration, authentication, database management, image storage, and system testing. The project demonstrates how a centralized digital platform can simplify laundry submission, maintain cloth-related records, and improve complaint handling.

Overall, Cloth Tracer fulfills its main objectives by providing an organized and user-friendly solution for students and laundry administrators. The system helps students manage their laundry records and report misplaced or damaged clothes, while administrators can manage submissions, complaints, and notifications efficiently. The project also provides a strong foundation for future improvements and real-world deployment.

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